# How Accurate Must Body Pose Be to Help? A Controlled Study of LiDAR-Recovered Pose for Pedestrian Trajectory Forecasting

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***Abstract*—Body pose is an intuitively attractive input for pedestrian trajectory forecasting: gait, stance and facing appear to encode intent that a position history does not. We test this on the Waymo Open Motion Dataset with SMPL poses recovered from LiDAR, and find no measurable benefit. The decisive comparison is not pose against no pose—which confounds body information with the capacity of an extra input branch—but correct pose against the same architecture fed *another pedestrian's* pose. On that control the effect is $+0.66\%$ ($p$=0.52) from scratch, $+0.88\%$ ($p$=0.18) after full-dataset pretraining, $-1.02\%$ ($p$=0.56) at an intermediate scale and $+0.04\%$ ($p$=0.93) at $43\times$ more training scenes, while the baseline improves $19\%$ from data alone; at the largest scale the data bound the content effect to $\pm 1.7\%$. A directed search over the integration code accepted one of $22$ candidates under a truncated proxy, and the pre-registered eight-seed confirmation that followed returned $-0.04\%$ ($p$=0.92). An oracle control rules out a dead pathway: the same branch, fed the agent's own future instead of its body, improves minADE by $34.8\%$. We then measure why pose does not. Corrupting that oracle with a controlled angular error gives the pathway a *detection floor*: the gain survives a median bearing error of $10.0^\circ$ ($p$=0.04) but is already indistinguishable from zero at $15.1^\circ$, where the ladder's minimum detectable effect is $6.7$ pp. SMPL body facing's median error, $12.3^\circ$, falls inside that bracket—but $42\%$ of pedestrians sit above it, and the median is largely inherited from the trajectory, since the pipeline snaps body orientation to the direction of travel for movers. For pedestrians stationary at prediction time, whose facing their velocity does not supply, facing reaches $41.2^\circ$, where the fitted decay predicts under $0.4\%$ against a branch-capacity cost of $\sim 1.6\%$. The floor is cheap to measure before a modality is integrated. Corroborating this, the tracker's box heading estimates the same quantity more accurately ($25.2^\circ$) and pose does not beat it in any range, occlusion or heading-jitter stratum. Finally, aggregate MPJPE is the wrong acceptance metric for pose in a perception stack: scored per joint against a mean-pose baseline, the head's MPJPE looks *better* than either wrist while it carries a tenth of their skill.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, negative results**

## I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes, and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose.

This paper reports that, on the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy—and, more usefully for practitioners, identifies a specific reason that can be checked quantitatively, and cheaply, before a modality is integrated at all.

We make three contributions.

**A controlled null.** We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch, and it is the comparison a practitioner should demand before adopting any auxiliary modality. The control is null at three training scales; across six pose encodings (6D joint rotations, 3D joint positions, root-orientation-free, agent-frame root, constant-pose and eight frame-invariant descriptors) and two encoder architectures (GRU and cross-attention); under two integration styles, a directed architecture search, and a pre-registered eight-seed confirmation on a held-out split. An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by $34.8\%$ through the same branch—so the null is a property of pose, not of the plumbing.

**A detection floor.** We then measure how accurate a side channel must be before this pathway converts it into accuracy at all, by corrupting that oracle with a controlled angular error. The gain decays by roughly half for every 3–8° of bearing error: it is still significant at $10.0^\circ$ and is indistinguishable from zero by $15.1^\circ$. SMPL body facing's median error, $12.3^\circ$, lands inside that bracket—but the median is inherited, because the pipeline snaps body orientation to the direction of travel for moving pedestrians, and $42\%$ of pedestrians sit above the bracket anyway. For the stationary ones—where facing is precisely the cue velocity cannot supply—it reaches $41.2^\circ$, where every seed's fitted decay predicts a gain under $0.4\%$ against a branch cost of $\sim 1.6\%$. The floor can be measured before any modality is integrated.

**A measurement caveat.** Aggregate MPJPE cannot distin-

guish a pose estimator that tracks a joint imprecisely from one that ignores the input and emits its prior. Scoring each joint against an always-predict-the-mean-pose baseline, MPJPE is uninformative about which joints are actually recovered ($\rho = -0.11$ across joints, n.s.) and inverts the ordering for the head. Accepting a pose estimator into an AV stack on a single error number risks importing a prior.

We also report, as a cautionary note, that our first attempt at this study was invalidated by an export bug that silently substituted other pedestrians' bodies—a failure any pose pipeline keyed by object identity can reproduce.

## II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers; we build on MTR [1], [2], which is representative and strong. Human mesh recovery from LiDAR is maturing [3], with motion priors used to complete sparse observations [4]–[6]. Intent-focused work on pedestrian datasets [7], [8] targets crossing decisions rather than metric trajectories.

**Relation to prior positive results.** Pose conditioning has reported real gains; our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [9] lowers ADE on the synthetic JTA dataset from $0.99$ to $0.89$ with *ground-truth* 3D pose, and—the harder case for us—from $0.40$ to $0.36$ on JRDB with *estimated* pose. Social-Pose [10] reports about $25\%$ on JTA ($1.20 \rightarrow 0.90$) and on JRDB. Salzmann et al. [11] report $\approx 10\%$ from onboard-estimated keypoints at ranges up to 7 m, specifically for agents with little position history, which is also our best hypothesis.

Le et al. [12] reach the opposite conclusion on JRDB and, in doing so, say the most useful thing about ours: they localise the gain to *lower-body* 3D keypoints ($13\%$ ADE, a further 1–$4\%$ from derived biomechanical cues). Their cue is our weakest channel. Of the twelve limb joints in Table IV the knees and ankles that carry gait sit at the bottom of the range—SKILL $+0.46$ to $+0.49$, against $+0.63$ and $+0.64$ for the elbows—and what leg motion we do recover is damped: predicted knee and ankle spread is 84–$91\%$ of ground truth, where elbows and wrists reach 96–$99\%$. A panoramic camera at a few metres resolves a swinging leg; a few dozen LiDAR returns at $40$ m recover its mean position and lose much of its amplitude. So the two results do not conflict—theirs is what this cue is worth above the detection floor of Sec. V-A, ours is what it is worth below.

The most informative comparison is within Social-Pose itself. Alongside JTA it evaluates on Pedestrians and Cyclists in Road Traffic, the one vehicle-captured dataset there, and—with *ground-truth* 3D pose—ADE moves from $0.58$ to $0.57$ m, by $0.01$ m or less for all five architectures tested, against the $\approx 25\%$ the same encoder delivers on JTA. The authors attribute this to the shorter horizon and the absence of interacting neighbours; our measurements suggest an additional reason that travels with the sensor rather than the horizon. Either way, the closest published setting to ours already reports approximately our result.

Two properties separate the remaining positive settings from an AV stack, and both follow from the mechanism in Sec. V. *Range and sensor:* JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians are observed from 2 to 75 m and often return a few dozen LiDAR points. *The baseline:* those predictors consume positions, not a tracker's box heading, so body facing is new information there and partly duplicated here. On controls, none reports a shuffled-pose arm—though the confound differs by design. Social-Transmotion masks modalities within one network, so no extra parameters are involved there, whereas a dedicated pose branch pays the capacity cost we measure ($+1.7\%$ minADE)—the same order as the smaller reported gains.

## III. Setup

### A. Data and a data-integrity failure

We use WOMD [13] with SMPL [14] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [15], which fits per-frame bodies with LiDAR-HMR [3] and completes them using AMASS-trained motion priors [4]–[6]. Poses cover the $11$ observed frames; the prediction horizon is $8$ s at $10$ Hz. The branch is not starved: every predicted agent carries at least one valid pose frame, and $86.7\%$ of observed frames carry one, so sparsity of pose is not an explanation for anything that follows.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than $90°$, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below $0.3$ m/s.

Our first pass at this study was invalid. The export keyed pose clips by a scene-*local* object identifier, so clips collided across scenes and nearly every training sequence carried another pedestrian's body. A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from $41.5°$ to $11.4°$ and raised the gait–speed correlation from $r = 0.26$ to $0.58$. We report it because the failure is silent—training converges, losses look normal—and identity-keyed joins are common in pose pipelines. Every number here is from the corrected export.

### B. Arms and the content test

Three arms share one architecture (Fig. 1). *No-pose* receives trajectories and map. *Pose* adds the pedestrian's own SMPL sequence. *Shuffled* receives another pedestrian's sequence, with identical parameter count, optimiser and input dimensionality. A fourth arm, *oracle*, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch and its masks untouched; it shows the pathway can carry information at all (Sec. IV-D).

Pose versus no-pose confounds information with branch capacity; pose versus shuffled isolates content. We call the latter

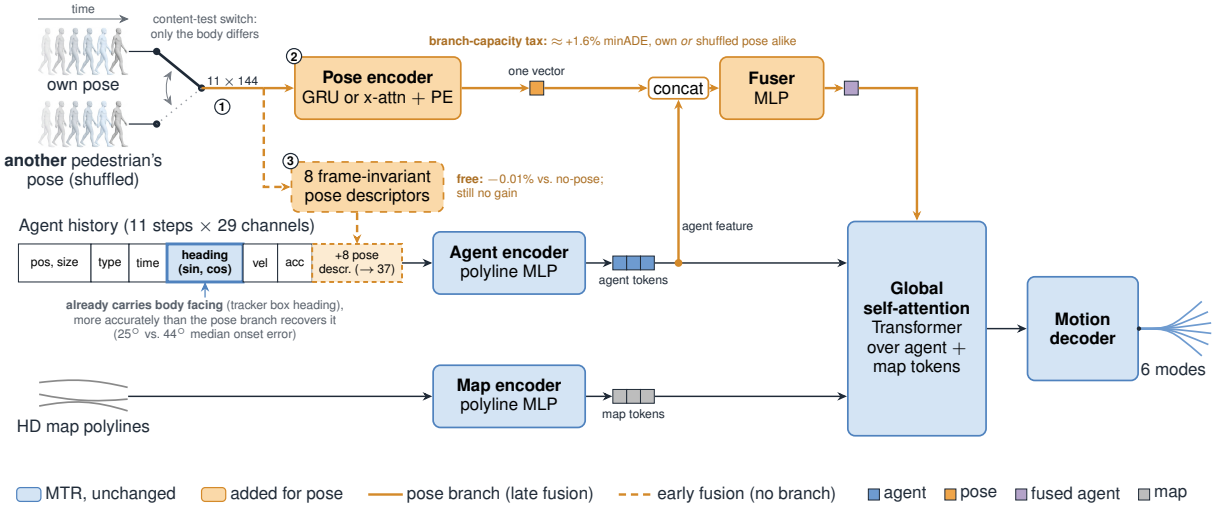



Fig. 1. The two arms differ only in *whose* body enters the model. The content-test switch (1) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed; everything downstream is shared. Blue: MTR, unchanged; orange: added for pose. The agent-history strip shows that the tracker's box heading is *already* an input. Solid orange (2): the pose branch (late fusion), which costs ≈1.6% minADE whether it carries the right body or the wrong one; dashed (3): early fusion, which removes that cost and still yields no benefit—and which, fed an uncorrupted oracle, fails to exploit it on two seeds in three (Sec. V-C).

the *content test* and treat it as decisive. Relative differences are $(A - B)/B$; negative favours $A$.

Inference is over *seeds*, which is the unit the treatment is applied to: every comparison between trained arms is a paired $t$-test on per-seed differences. The two exceptions are flagged where they occur—per-pedestrian subpopulation tests and the per-joint intervals of Table IV, which resample pedestrians. We also report a hierarchical bootstrap that resamples seeds and then pedestrians, and it agrees; we do not quote a percentile bootstrap $p$-value, because with five seeds whose differences share a sign it returns the resolution floor $1/N_{\text{boot}}$ rather than a calibrated tail probability, and a pedestrian-only bootstrap ignores between-seed variance entirely.

### C. Protocol

Every arm trains for 30 epochs with AdamW (batch 2 per GPU, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25; the decoder predicts $K{=}6$ modes. The learning rate is chosen per rung from a grid run separately for each arm: $1{\times}10^{-4}$ at 579 scenes, $5{\times}10^{-5}$ at 5k and $2.5{\times}10^{-5}$ at 25k, where every arm peaks at the same interior optimum. Validation is fixed at 5,171 scenes and 16,486 pedestrians; the directed search of Sec. IV-C splits it once into a development half (2,617 scenes, 8,445 agents) and a held-out half (2,554 scenes, 8,041 agents). Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP and gated miss rate need the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER RUNG BY A GRID RUN SEPARATELY FOR EVERY ARM; ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS; $p$-VALUES ARE PAIRED $t$-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

| Training scenes | 579 | 5k | 25k |
|---|---|---|---|
| Pedestrians | 6,348 | 31,302 | 142,807 |
| Seeds | 5 | 3 | 5 |
| *minADE* (m) | | | |
| No-pose | 0.4599 | 0.4176 | 0.3716 |
| Pose | 0.4620 | 0.4138 | 0.3778 |
| Shuffled | 0.4589 | 0.4181 | 0.3777 |
| Pose vs. no-pose | +0.45% | −0.90% | +1.66%* |
| **Content test** | +0.66% | −1.02% | +0.04% |
| $p$ (content) | 0.52 | 0.56 | 0.93 |
| *minFDE* (m) | | | |
| No-pose | 0.9922 | 0.8992 | 0.8049 |
| Pose | 0.9824 | 0.8875 | 0.8131 |
| Shuffled | 0.9780 | 0.8900 | 0.8104 |
| Pose vs. no-pose | −0.99% | −1.30% | +1.02%† |
| **Content test** | +0.45% | −0.28% | +0.33% |
| $p$ (content) | 0.63 | 0.83 | 0.61 |

*$t(4){=}10.3$, $p{=}5{\times}10^{-4}$; the shuffled arm pays the same cost, +1.63%, $t(4){=}2.8$, $p{=}0.048$.
†$t(4){=}6.1$, $p{=}0.004$.

## IV. RESULTS

### A. The null, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%; the content test is null at every rung, in minADE and in minFDE alike, and in miss rate at 2 m, where the content test is +2.3% at 25k ($p{=}0.21$) and +4.5% fine-tuned ($p{=}0.19$)—null, and with its sign against pose on all three metrics. The rungs are not equally sensitive, and we state what each one could have seen: at 80% power

TABLE II
CLEAN CORE AT 579 SCENES, $n$=5 SEEDS, BOTH REGIMES. THE SHUFFLED ARM HAS THE LOWEST MEAN IN BOTH: ANOTHER PEDESTRIAN'S BODY IS NOMINALLY THE BEST INPUT.

| Arm | Per-seed minADE | Mean ± sd | Content |
|---|---|---|---|
| *From scratch* | | | |
| No-pose | .458 .443 .453 .495 .450 | $0.4599 \pm 0.0206$ | — |
| Shuffled | .454 .467 .451 .455 .467 | $0.4589 \pm 0.0075$ | — |
| Pose | .471 .456 .453 .457 .474 | $0.4620 \pm 0.0094$ | $+0.66\%$ |
| *Fine-tuned on a full-WOMD backbone* | | | |
| No-pose | .372 .366 .367 .373 .368 | $0.3692 \pm 0.0031$ | — |
| Shuffled | .364 .363 .364 .374 .373 | $0.3676 \pm 0.0055$ | — |
| Pose | .364 .372 .371 .374 .373 | $0.3708 \pm 0.0040$ | $+0.88\%$ |

the minimum detectable content effect is $3.5\%$ at 579 scenes ($n$=5), $7.6\%$ at 5k ($n$=3) and $2.2\%$ at 25k ($n$=5). The 5k rung is therefore weak evidence on its own and we do not count it as an independent null; the weight falls on 25k and on the eight-seed confirmation of Sec. IV-C. At 25k the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$, so at the largest scale we train the data exclude a content effect beyond roughly $1.7\%$ in either direction—a bound tighter than the $1.6\%$ cost of the branch that would carry it. The pedestrian-resampling bootstrap gives a visibly narrower interval ($[-0.0040, +0.0047]$ m, $\approx \pm 1.2\%$); with five clusters it undercovers, and we quote the wider seed-level bound throughout.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning ($n$=5), the content test is $+0.88\%$ ($p$=0.18), again with its sign against pose (Table II).

Seed count decides what Table II appears to say. On the first three seeds the same runs read as *pose harms* from scratch ($+1.85\%$) and *shuffled is best* after fine-tuning ($-1.28\%$); at $n$=5 both shrink roughly fourfold. Both three-seed effects are significant under a pedestrian-resampling bootstrap ($p$=0.037, $p$=0.016) and neither is under a paired $t$ over the same seeds ($p$=0.18, $p$=0.12): three seeds do not resolve a $1\%$ effect here, and a bootstrap over the wrong unit will not say so.

### B. Two effects that are real, and are not pose

**A branch-capacity tax.** At 25k both pose arms are worse than no-pose ($+1.66\%$, $p=5\times10^{-4}$; $+1.63\%$, $p$=0.048) and indistinguishable from each other. An extra input branch costs $\sim 1.6\%$ whether it carries the right body or the wrong one.

This is an operating-point statement, and the operating point matters: it is measured at each run's best validation epoch. Models with wider inputs overfit sooner, so the sign of a pose-versus-no-pose comparison is not stable under a fixed late-epoch rule, where the no-pose arm has degraded further by then ($+8.0\%$ in favour of pose at epoch 30, $n$=3; $+0.9\%$ at $n$=8, both n.s. and neither resolved at these seed counts). What is stable is that the comparison moves with the stopping rule, so we never compare architecturally different arms at a fixed late epoch. The *content* test—whose two arms have identical input dimensionality, and for which this asymmetry does not arise—is null under either rule. Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels removes the cost ($-0.01\%$ versus no-pose, $p$=0.99; $-1.67\%$ versus the branch, $p$=0.12 at $n$=3) without creating a benefit: pose can be carried for free, and still does not pay. That freedom turns out to be partly illusory—Sec. V-C shows this pathway fails to exploit even an uncorrupted oracle on two seeds in three, so a channel it costs nothing to carry is also a channel it largely does not read. We quote the seed-level $p$ here for consistency with the rest of the paper; a pedestrian-resampling bootstrap makes the branch contrast look decisive ($p$=0.006), which is exactly the unit error Sec. III warns against.

**Tuning dominates the treatment.** At 5k the learning rate inherited from the smaller rung made pose look catastrophic ($+37\%$); with per-arm tuning, $-0.90\%$, n.s. Across the 25k grid the learning rate moves minADE by up to $16.7\%$ against a content signal of $\sim 0.05\%$: a pose-versus-baseline comparison that does not tune each arm separately is a claim about the learning rate.

### C. Search and confirmation

Hand-designed integrations sample a small space, so we ran a directed search in which a coding agent rewrote the pose-integration code against a frozen training repository, scoring on a development split and never seeing the held-out split. Round one scored 13 candidates and accepted none. Round two produced one acceptance: a stationary-gated body-facing plus foot-contact descriptor that beat the incumbent on $5/5$ seeds and reached $-1.21\%$ content on the held-out split, clearing the pre-set $0.8\%$ acceptance floor—under a truncated 12-epoch proxy recipe.

Because that acceptance rested on a truncated recipe, we pre-registered a confirmation before running it: three arms, eight seeds, 30 epochs, primary endpoint the held-out split at the fixed last epoch. It returned $-0.04\%$, $t(7) = -0.11$, $p$=0.92. The proxy signal was a convergence-rate artifact—the arms differ in how fast they train, not in where they end—and a truncated proxy cannot settle a sub-$1\%$ effect. We report the acceptance rather than only the confirmation because the search behaved honestly: it found the best candidate the representation admits, and pre-registration is what removed it.

### D. The branch is not inert

A null measured through one pathway invites the obvious objection that the pathway is dead. The shuffled control does not dispose of it: it shows the model cannot distinguish its own body from a stranger's, which is equally consistent with a branch that transmits nothing at all. The branch-capacity tax shows the branch perturbs the output, not that it carries signal, and the fusion residual is zero-initialised, so a dead branch and an uninformative one begin at the same place.

We therefore ran the positive control. The *oracle* arm replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s — genuinely predictive by construction — keeping the branch, its parameter count, its input dimensionality and its per-frame validity masks

exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds the oracle arm reaches $0.2945$ against no-pose's $0.4514$ on the same seeds—$-34.8\%$, $t(2) = -81.4$, $p = 1.5 \times 10^{-4}$—and $-35.6\%$ against the shuffled arm. The oracle is of course not a usable model, since it reads the future; its only job is to calibrate the pathway. It does so emphatically: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given.

### *E. Where a benefit could still hide*

On the confirmation runs the content test is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, $-0.04\%$ at 8 s); null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration; and null for standing-then-moving pedestrians, the subpopulation built to favour it ($n$=1,809 pedestrians, $+0.38\%$, epoch-30 checkpoints on an expanded validation split; $p = 0.80$ by a pedestrian-resampling bootstrap, the one place we quote one, because this subpopulation is defined per pedestrian).

Across the roughly forty content tests this study reports, one reaches $p < 0.05$: in the stationary subpopulation of the heading ablation, correct pose is $+1.8\%$ *worse* than a stranger's ($n$=8, $p = 0.011$). At that multiplicity one such result is expected by chance, and its direction is against a pose mechanism rather than for one; we report it so that the released metrics contain no surprise.

## V. Why Pose Does Not Pay

### *A. A detection floor for auxiliary channels*

The content test bounds what pose is worth, and the oracle of Sec. IV-D shows the branch can carry information. Neither says how *much* information a side channel must carry before this pathway turns it into accuracy. We measured that directly.

Each rung of Table III feeds the branch the same direction-only oracle—the bearing of the agent's own future displacement at 3 and 8 s—corrupted by a fixed angular error, and is labelled by the median bearing error actually delivered—$0.6745\sigma$ by construction, since the corruption is a fixed per-agent rotation, and measured to within $0.7^\circ$ of it on 11,328 scored agents. Displacement magnitude is withheld so the threshold can be stated in degrees. The error is drawn once per pedestrian, not per epoch: a real estimator's error is a property of the observation, and resampling it would let training average it away.

The improvement decays with bearing error. It is large and significant when the bearing is exact ($-15.8\%$), at $5.0^\circ$ ($-7.6\%$, $p$=0.0004) and at $10.0^\circ$ ($-6.0\%$, $p$=0.040), and it is no longer distinguishable from zero at $15.1^\circ$ ($-1.4\%$, $p$=0.47) or anywhere above it. Fitting all eight rungs,

$$\Delta minADE(\theta) \;\approx\; -15.7\% \cdot \exp(-\theta/7.7^\circ), \qquad (1)$$

a half-life of $\approx 5^\circ$. The seeds do not pin these constants down: fitting each separately against its own no-pose run gives amplitudes of 12–20% and half-lives of 3.1–8.0°, so we read (1) as an order-of-magnitude law and state the floor as a *bracket*—the gain survives $10.0^\circ$ and is undetectable beyond $\approx 15^\circ$—rather than as a crossing point. Above the bracket the values scatter around zero, and the ladder cannot resolve anything below $\approx 6$ pp there, so the high rungs ($+2.7\%$ at $44^\circ$) are indistinguishable from zero and from the branch tax of Sec. IV-B; we read no sign change into them.

The two rungs bounding the bracket were taken to five seeds, which exposed a dependence worth stating. One seed's no-pose run is a $6.8\,\sigma$ outlier against the other four and is the shared denominator for that seed's whole column; retraining it gave a draw $0.023$ lower, more than the span of the other four, so run-to-run noise at fixed seed rivals the between-seed spread. Both draws are legitimate, so the bracket uses their mean. Retraining, averaging or dropping that seed all give $[10.0^\circ, 15.1^\circ]$; only the single original draw gives $[5.0^\circ, 10.0^\circ]$. The lower edge is the least robust number here.

### *B. Where body facing sits relative to the floor*

The floor is calibrated in displacement-bearing error over every predicted pedestrian, so we measure body facing against that target, on that population: median $12.3^\circ$ at $+3$ s ($n$=5,824), $15.1^\circ$ at $+8$ s ($n$=6,206). Read naively that is *not* below the floor—each seed's fit at $12.3^\circ$ predicts 1.3–4.0%, at or above the $\sim 1.6\%$ branch tax—a comparison across rungs, since that tax is resolved only at 25k, while at the ladder's own 579-scene rung the same contrast is $+0.61\%$ with a 95% interval of $\pm 6$ pp. Three properties of that median remove the apparent margin.

*Most of it is the trajectory restated.* $95.5\%$ of the scored agents move, and for a mover the pipeline snaps root orientation to the direction of travel whenever the raw estimate deviates by more than $90^\circ$ (Sec. III). Movers sit at $11.9^\circ$; the 262 stationary agents—the only ones whose facing their velocity does not already imply—sit at $41.2^\circ$, where the per-seed fits predict 0.002–0.32%. The pooled median largely measures the snapping.

*A rung is a uniform corruption; SMPL is a heavy tail.* Each rung applies its angle to *every* agent, so a channel is comparable to a rung only if its whole distribution sits there. SMPL's does not: $5.5^\circ$ at the first quartile, $23.9^\circ$ at the third, $45.9^\circ$ at the ninetieth percentile. $42.2\%$ of agents sit above the bracket's upper edge at $+3$ s and $50.0\%$ at $+8$ s, and $57.5\%$ are above its lower edge. Reading an exponential decay at a median flatters the channel for the same reason, since the badly estimated half contributes almost none of the gain.

*It is a noisier copy of an input the model already has.* The tracker's box heading estimates the same bearing at $3.9^\circ$ ($+3$ s) and $7.1^\circ$ ($+8$ s), beats SMPL at every percentile we measured, and clears the bracket for all but $14.2\%$ of agents—even though we calibrated the SMPL forward axis *against* the box heading first. The two differ by $10.0^\circ$ in median.

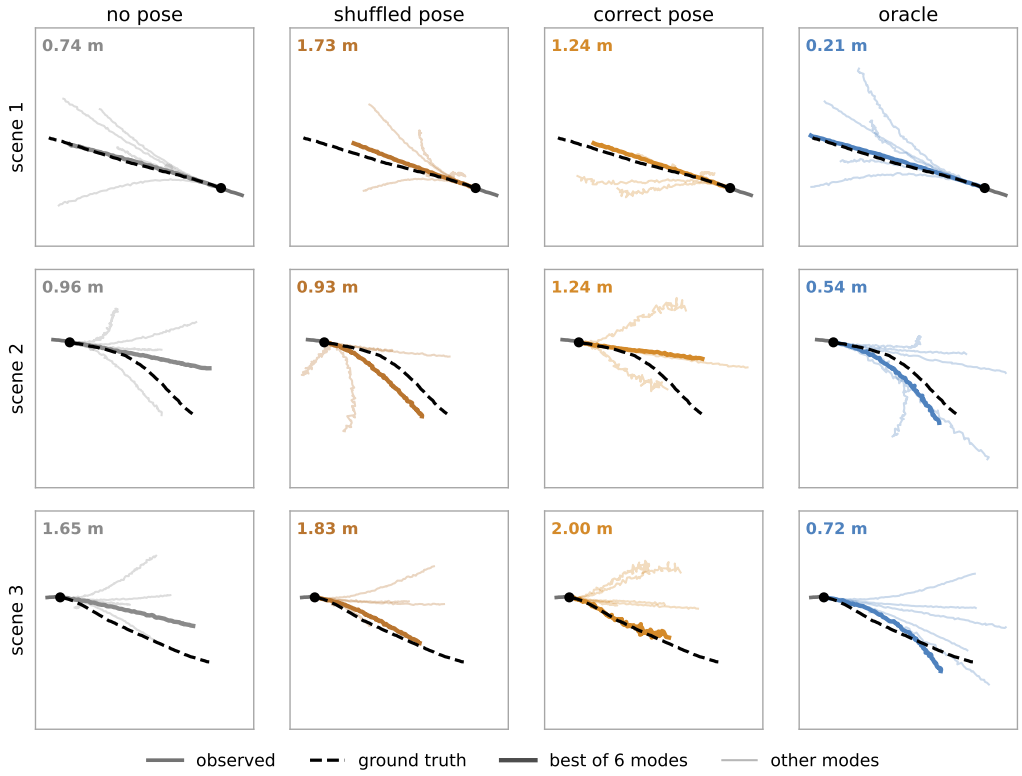



Fig. 2. Qualitative comparison of the four arms on the same three agents (seed 101, epoch 30, the epoch for which full trajectory dumps were retained). Scenes are picked by percentile of the *no-pose* arm's minADE (60th, 76th, 92nd) among agents with a full 8 s future and more than 3 m of travel—selection is on the baseline alone, never on the treatment. Thin lines are the six predicted modes, thick the best of them, dashed the ground truth, grey the observed past. Fed the agent's own future bearing, the oracle bends its modes onto the true path through the *same* branch that carries pose; correct pose is not consistently distinguishable from a stranger's pose or from no pose. Per-panel values are single-seed illustrations: across seeds the oracle's advantage is stable ($-34$ to $-40\%$), whereas the content difference at this fixed epoch is not ($-16.2$, $-8.0$, $-0.2\%$ for seeds 101/202/303) and is null at the best-epoch operating point used for inference (Table I).

So facing clears the bracket mainly for the agents whose facing their own velocity already supplies, and sits above it for two in five—most plainly for the stationary pedestrians the cue was introduced to serve.

**The onset task agrees.** On the one case where velocity carries no directional information—which way a stationary pedestrian steps off—SMPL facing predicts onset direction at roughly twice chance over 19,939 validation pedestrians: median error $44.2°$, $50.6\%$ within $45°$ (chance $90°$, $25\%$). Informative, consistent with the $41.2°$ above, and not an artifact of our axis fitting: the box heading estimates the same quantity at $25.2°$, and paired on the same events SMPL is $7.3°$ worse in median (Wilcoxon $p{=}2.6{\times}10^{-43}$). Nor does pose take over where perception degrades—over 1,705 onset events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and *widens* with range, to $+32.9°$ at 41–75 m, because both derive from the same returns.

**What this does not establish.** On the stationary pedestrians the box heading is *also* outside the bracket—(1) at its $25.2°$ predicts $0.3\%$, per-seed $0.1$–$1.4\%$, under the branch tax on every seed—so pose does not lose to heading because heading is accurate enough to pay through this pathway; through a late-fused branch neither accuracy would. Pooled, the box heading's $3.9°$ is well inside the bracket, but it reaches the model as a native per-timestep input rather than through a

TABLE III

DETECTION FLOOR FOR THE POSE BRANCH. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH RUNG LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED; PAIRED $t$ AGAINST EACH SEED'S OWN NO-POSE RUN, NEGATIVE FAVOURS THE ORACLE. THE TWO RUNGS BOUNDING THE BRACKET WERE EXTENDED TO FIVE SEEDS. MDE IS THE MINIMUM EFFECT A RUNG COULD DETECT AT 80% POWER GIVEN ITS REALISED SPREAD, SO THE FLOOR IS REPORTED AS THE BRACKET $[10.0°, 15.1°]$ RATHER THAN A CROSSING POINT. TWO IN FIVE PEDESTRIANS HAVE A FACING ERROR ABOVE THAT BRACKET, AND THE STATIONARY ONES THE CUE WAS INTRODUCED TO SERVE SIT FAR ABOVE IT (SEC. V-B).

| Bearing error | $n$ | ΔminADE | $p$ | 95% CI | MDE |
|---|---|---|---|---|---|
| $0.0°$ | 3 | $-15.8\%$ | 0.021 | $[-25.6, -5.9]$ | 12.3 |
| $5.0°$ | 3 | $-7.6\%$ | 0.0004 | $[-8.2, -6.9]$ | 0.8 |
| $10.0°$ | 5 | $-6.0\%$ | 0.040 | $[-11.6, -0.5]$ | 7.4 |
| $15.1°$ | 5 | $-1.4\%$ | 0.472 | $[-6.5, +3.6]$ | 6.7 |
| $24.9°$ | 3 | $-0.1\%$ | 0.954 | $[-5.4, +5.2]$ | 6.6 |
| $35.7°$ | 3 | $+0.6\%$ | 0.624 | $[-3.8, +5.0]$ | 5.5 |
| $44.4°$ | 3 | $+2.7\%$ | 0.252 | $[-4.6, +10.0]$ | 9.1 |
| $60.1°$ | 3 | $+2.7\%$ | 0.237 | $[-4.2, +9.5]$ | 8.5 |

branch, and we have not laddered that channel. The floor is a property of the pathway, not of every way a cue can enter a model—and Sec. V-C ladders the one alternative we could test, finding it converts the cue even less reliably.

We tested the implied counterfactual directly. Removing the heading channel entirely and repeating the content test changes it by $-0.6$ pp in the stationary subpopulation ($n$=8, $p$=0.65, CI $[-3.5, +2.3]$): no effect. That is consistent with (1) for a channel never inside the bracket, though equally consistent with pose carrying nothing the trajectory does not already supply; it rules out redundancy as the *cause* rather than confirming the floor. At $n$=3 the same comparison read $-4.1$ pp with all three seeds agreeing in sign—the third multi-percent effect here to dissolve under more seeds.

### C. The cheaper pathway has no floor to measure

Early fusion removes the branch tax (Sec. IV-B), which makes it the pathway worth recommending—so we ran the same ladder through it, carrying the oracle in the per-timestep agent channels instead of the branch. At a given rung the six channels equal the branch's six inputs bit for bit, so the rungs keep their measured-error labels.

No rung is significant, *including the uncorrupted one*: $-4.5\%$ at $\theta$=0 ($p$=0.44, MDE 25.5), $-1.0\%$ at $5.0^\circ$ ($p$=0.53) and $+0.2\%$ at $10.0^\circ$ ($p$=0.94), at $n$=3 throughout. That is not a lower floor; it is the absence of a curve on which to locate one. At $\theta$=0 the channel is the agent's own future bearing, and the side branch converts it on all three seeds ($-16.5\%$, $-19.3\%$, $-11.5\%$); early fusion converts it on one ($-14.0\%$, against $+0.7\%$ and $-0.3\%$). The two failing seeds sit at their no-pose level from $\theta$=0 onward, so the cue reaches the input and is not used.

This reframes the free-carriage result of Sec. IV-B. "No tax, no benefit" admits two readings: that early fusion integrates pose at no capacity cost, or that it largely ignores the appended channels, in which case there is nothing to cost and nothing to gain. A perfect channel separates them—the first predicts a large gain, the second none—and we measure none on two seeds in three. We therefore withdraw the suggestion that a cue failing through the branch might still pay if fused early: where we could test it, the cheaper integration was the less reliable one. Caveats: $n$=3, and the free-carriage result was measured at 25k scenes while this ladder runs at 579.

### D. The complementary cue is barely measured

Gaze precedes locomotion, so head yaw relative to the torso is a plausible *independent* cue—and it is very nearly not recoverable here. In the fitted meshes head-versus-torso yaw has a standard deviation of $3.8^\circ$, while against the ground-truth 3D keypoints released with Waymo Perception [16], [17] ($n$=8,234) the real quantity varies $19.2^\circ$: a fivefold under-dispersion, so the estimator is largely emitting its prior. Consistently, head yaw carries no usable signal for onset direction ($r = -0.03$, n.s.), and the head is the one joint in Table IV whose interval includes zero. At 20–40 m the head is a near-spherical handful of points with little geometry to constrain yaw.

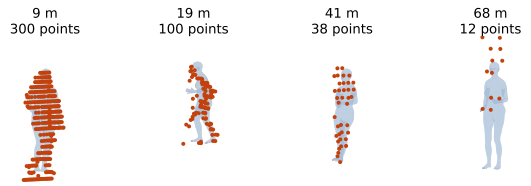



Fig. 3. What the pose estimator has to work with. Raw per-pedestrian LiDAR returns (orange) and the SMPL mesh fitted to them (blue), at four ranges and a common metric scale. Each panel is a frame whose return count equals the median for its range band over the validation segments: 300, 100, 38 and 12 points. Past ≈40 m a pedestrian is a few dozen returns, mostly on head and torso, while the mesh over them still has fully articulated limbs: at that range the articulation is largely the AMASS prior. That applies to the far panels only—scored against ground-truth keypoints across all ranges, every limb joint carries real skill, and it is the *head* the pipeline emits from its prior (Table IV). Sparsity at range is therefore one contributor to the facing error of Sec. V-B, where the pose-versus-heading gap widens to $+32.9^\circ$ at 41–75 m, not its whole explanation.

TABLE IV
PER-JOINT RECOVERABILITY AGAINST GROUND-TRUTH KEYPOINTS. MP IS THE ERROR OF ALWAYS PREDICTING THE MEAN POSE (EQUAL TO THE JOINT'S GROUND-TRUTH SPREAD BY CONSTRUCTION); SKILL $= 1 - \mathrm{MPJPE}/\mathrm{MP}$. THE HEAD'S MPJPE BEATS BOTH WRISTS WHILE CARRYING A TENTH OF THEIR SKILL. INTERVALS RESAMPLE PEDESTRIANS ($n$=534), NOT FRAMES, SINCE FRAMES WITHIN A TRACK ARE STRONGLY CORRELATED. EVERY LIMB INTERVAL CLEARS ZERO BY A WIDE MARGIN (LOWEST BOUND $+0.43$); THE HEAD'S IS $[-0.003, +0.089]$, THE ONE INTERVAL THAT CONTAINS IT.

| Joint | MPJPE (mm) | MP (mm) | SKILL |
|---|---|---|---|
| Elbows | 91.2 / 91.5 | 252.9 / 248.9 | +0.64 / +0.63 |
| Hips | 39.8 | 97.9 | +0.59 |
| Wrists | 118.3 / 125.4 | 271.1 / 276.4 | +0.56 / +0.55 |
| Shoulders | 87.3 / 89.0 | 171.5 / 164.0 | +0.49 / +0.46 |
| Ankles | 114.0 / 116.6 | 221.6 / 223.8 | +0.49 / +0.48 |
| Knees | 78.4 / 81.5 | 150.8 / 151.2 | +0.48 / +0.46 |
| **Head** | **111.6** | **116.9** | **+0.045** |

### E. MPJPE is the wrong acceptance metric

A single mean error cannot distinguish an estimator that tracks a joint imprecisely from one that emits its prior. We therefore score each joint against the strongest trivial competitor, always predicting the mean root-aligned pose: $\mathrm{SKILL}_j = 1 - \mathrm{MPJPE}_j/\mathrm{MP}_j$, where values at or below zero mean the estimator contributes nothing at that joint.

Evaluated on Waymo Perception against ground-truth keypoints (1,789 frames, 534 pedestrians), every limb joint scores $+0.457$ to $+0.639$ (Table IV). The head scores $+0.045$—yet its MPJPE (111.6 mm) is *better* than either wrist (118.3/125.4 mm). Its error is flattering not because it is well estimated but because it barely improves on its own mean-pose baseline (116.9 mm), whereas the hips—whose ground-truth spread is *smaller* at 97.9 mm—are predicted at 39.8 mm. Across the thirteen joints MPJPE and skill are essentially uncorrelated (Spearman $\rho = -0.11$, $p$=0.72): the aggregate is not merely a noisy proxy for information content but carries

almost none of it, and for the head it actively inverts the ordering. We recommend reporting per-joint skill alongside it whenever a pose estimator is accepted into a downstream stack.

### F. A general condition

Two results locate where pose could have helped. At 2 s, minADE is $0.091$ m—about $3\%$ of the displacement—so little error is left for an extra input to remove. This is a property of the forecaster's own residual, not of constant velocity: single-mode constant-velocity extrapolation [18] gives $0.310$ m at 2 s on the same set, so short-horizon forecasting is *not* CV-saturated (a comparison that favours the model, since minADE takes the best of $K{=}6$ modes). Saturation here means the baseline's own remaining error is a few percent of the displacement. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose; stratifying by how badly constant velocity fails still yields no effect.

In general form: *an auxiliary modality improves a forecast only where the baseline is neither saturated nor dominated by intent the modality cannot observe, and where the modality's own measurement error leaves it above the integration pathway's detection floor.* All three clauses must hold. The third is the cheapest to check and the one most often skipped—and for pose on WOMD it is the one that fails: body facing is inside the bracket only for the agents whose facing their own velocity already supplies, and outside it for the majority.

## VI. What This Means in Practice

For an AV stack, three consequences follow. First, before integrating an auxiliary modality, measure its error on the cue it is meant to carry and compare that against the floor of the pathway you intend to ship—an oracle ladder costs a handful of runs and no new sensor. That floor belongs to the pathway, not the modality: we laddered two, and the cheaper one was not the safer one (Sec. V-C). Second, the capacity cost of an extra branch is real ($\sim$1.6% here, at the best-epoch operating point) and is easily mistaken for a modality effect in either direction; early fusion avoids it, though on our evidence it does so partly by not reading the channel, and a shuffled-pose control separates the two. Third, do not accept a pose estimator on aggregate MPJPE alone: check per-joint skill, because the joints an aggregate flatters may be the ones carrying a prior.

**Limitations.** The largest rung we train is 25k scenes of the 299,220 exported, so the full rung is a compute limit rather than an annotation one; at eight seeds the minimum detectable content effect is $\approx$1.25% at 80% power, and the ladder's own rungs are far coarser ($n{=}3$, or 5 at the two rungs bounding the bracket, Table III). We test one forecaster, one dataset and one pose estimator. We measure the floor on two integration pathways rather than one, but two is not every way a cue can enter a model, and the early-fusion ladder is $n{=}3$ at the 579-scene rung. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale; on Waymo Perception, where raw imagery exists, an RGB estimator (HMR2.0) reaches $106.2$ mm root-aligned MPJPE against LiDAR-HMR's $87.6$ mm on the same 347 frames, so the obvious better-pose route is the worse one here. We do not model pose–pose interaction.

## VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD. The null is controlled against a shuffled-pose baseline and holds across scale, representation, integration, a directed search and a pre-registered confirmation; an oracle through the same branch rules out a dead pathway. It also has a quantitative mechanism: an oracle ladder shows this integration path converts a bearing cue into accuracy at $10^\circ$ but no longer by $15^\circ$, and body facing falls above that bracket for two in five pedestrians—by a factor of three for the stationary ones the cue was introduced to serve. Fusing the same cue early removes the branch's capacity cost but does not rescue it: that pathway fails to exploit even an uncorrupted bearing on two seeds in three, so both integration routes fail, for different and identifiable reasons. The lesson generalises past pose—measure a modality's error on the cue it is meant to carry, and compare it against the floor of the pathway you intend to use, before integrating it.

## References


[1] S. Shi *et al.*, "Motion transformer with global intention localization and local movement refinement," in *Advances in Neural Information Processing Systems*, S. Koyejo *et al.*, Eds., vol. 35. Curran Associates, Inc., 2022, pp. 6531–6543.

[2] ——, "MTR++: Multi-agent motion prediction with symmetric scene modeling and guided intention querying," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3955–3971, May 2024.

[3] B. Fan *et al.*, "LiDAR-HMR: 3D human mesh recovery from LiDAR," *arXiv preprint arXiv:2311.11971*, 2023.

[4] D. Rempe *et al.*, "HuMoR: 3D human motion model for robust pose estimation," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 11 468–11 479.

[5] C. He *et al.*, "NeMF: Neural motion fields for kinematic animation," in *Advances in Neural Information Processing Systems*, vol. 35, 2022.

[6] N. Mahmood *et al.*, "AMASS: Archive of motion capture as surface shapes," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, Oct. 2019, pp. 5441–5450.

[7] A. Rasouli *et al.*, "PIE: A large-scale dataset and models for pedestrian intention estimation and trajectory prediction," in *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6261–6270.

[8] Z. Fang and A. M. López, "Is the pedestrian going to cross? answering by 2d pose estimation," in *2018 IEEE Intelligent Vehicles Symposium (IV)*, 2018, pp. 1271–1276.

[9] S. Saadatnejad *et al.*, "Social-transmotion: Promptable human trajectory prediction," in *International Conference on Learning Representations*, 2024.

[10] Y. Gao, S. Saadatnejad, and A. Alahi, "Social-pose: Enhancing trajectory prediction with human body pose," *IEEE Transactions on Intelligent Transportation Systems*, 2025, arXiv:2507.22742.

[11] T. Salzmann *et al.*, "Robots that can see: Leveraging human pose for trajectory prediction," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7090–7097, Nov. 2023.

[12] N. Le, D. Song, and X. Xiao, "Legs over arms: On the predictive value of lower-body pose for human trajectory prediction from egocentric robot perception," *arXiv preprint*, 2026, arXiv:2602.09076.

[13] S. Ettinger *et al.*, "Large scale interactive motion forecasting for autonomous driving : The waymo open motion dataset," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, Oct. 2021, pp. 9690–9699.

[14] M. Loper *et al.*, "SMPL: a skinned multi-person linear model," *ACM Transactions on Graphics*, vol. 34, no. 6, pp. 1–16, Oct. 2015.
[15] G. Zhu *et al.*, "Waymo-3DSkelMo: A multi-agent 3D skeletal motion dataset for pedestrian interaction modeling in autonomous driving," in *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025, pp. 13 184–13 190.
[16] P. Sun *et al.*, "Scalability in perception for autonomous driving: Waymo open dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2446–2454.
[17] J. Zheng *et al.*, "Multi-modal 3D human pose estimation with 2D weak supervision in autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2022, pp. 4478–4487.
[18] C. Schöller *et al.*, "What the constant velocity model can teach us about pedestrian motion prediction," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 1696–1703, 2020.